

A Public-Data Bayesian Framework for External Wildfire Exposure Ranking of Transmission-Line Segments

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Abstract

Linear energy infrastructure increasingly requires segment-level prioritization under wildfire exposure, but many asset-condition and outage datasets are not public. This paper presents a reproducible public-data framework for external wildfire exposure ranking of California transmission-line segments. The lines are treated as receptor assets rather than ignition sources. Using California Energy Commission transmission lines, CAL FIRE FRAP fire perimeters and cause fields, gridMET weather summaries, LANDFIRE LF2024 fuel and canopy layers, and USGS 3DEP terrain, the workflow constructs a 2017–2023 segment-year panel for the Northern Sierra and Southern Cascades. The main validation label excludes CAL FIRE CAUSE = 11, Electrical Power, to reduce conflation between external fire exposure and electrical-power fire cause. A Bayesian logistic exposure model is compared with a deterministic public physics score and machine-learning baselines under a 2022–2023 temporal holdout. The Bayesian and regularized logistic models have similar discrimination, while the Bayesian model provides posterior exceedance probabilities and posterior top-decile rank probabilities for decision-facing triage. The framework is not an ignition, equipment-failure, outage, or responsibility model.

1 Introduction

Wildfire exposure increasingly affects linear infrastructure systems. For transmission corridors, planning and inspection decisions are often made at the line-segment scale, before the exact location and severity of future fires are known. Screening-level tools therefore need to prioritize segments using weather, terrain, fuel, and historical fire information while keeping the target definition clear.

Powerline-wildfire research can be separated into two related but distinct directions. One direction studies grid-to-fire processes, such as powerline-caused ignition. A second direction studies fire-to-grid exposure, damage, and service disruption. This paper addresses only a limited part of the second direction: external wildfire exposure ranking for transmission-line segments treated as receptor assets. It does not estimate powerline-caused ignition probability, equipment failure, outage probability, damage, or legal responsibility.

The public-data constraint is central to the design. Private utility datasets such as relay operations, outage logs, conductor type, asset age, inspection records, vegetation-management histories, and equipment condition are not used. The resulting target is therefore a screening-level external exposure ranking, not a full operational risk model.

Existing deterministic hazard scores and machine-learning classifiers commonly return a single score or probability. Rare-disaster triage also needs uncertainty-aware quantities that answer decision questions directly, such as whether a segment exceeds a probability threshold or whether it

plausibly belongs in the top decile of a ranked list. This motivates a Bayesian decision layer on top of public fuel, weather, terrain, and historical exposure covariates.

The contributions are:

1. a reproducible public-data segment-year exposure panel for California transmission-line segments;
2. cause-screened external wildfire exposure labels that exclude electrical-power fire causes from the main validation target;
3. public fuel, weather, and terrain covariate construction for segment-level ranking;
4. Bayesian posterior exceedance and posterior top-decile rank probabilities as decision-facing outputs;
5. temporal-holdout comparison with deterministic and machine-learning baselines; and
6. an initial Stan MCMC check of the Laplace posterior prototype.

Figure 1 summarizes this positioning relative to deterministic scores and flexible ML baselines.

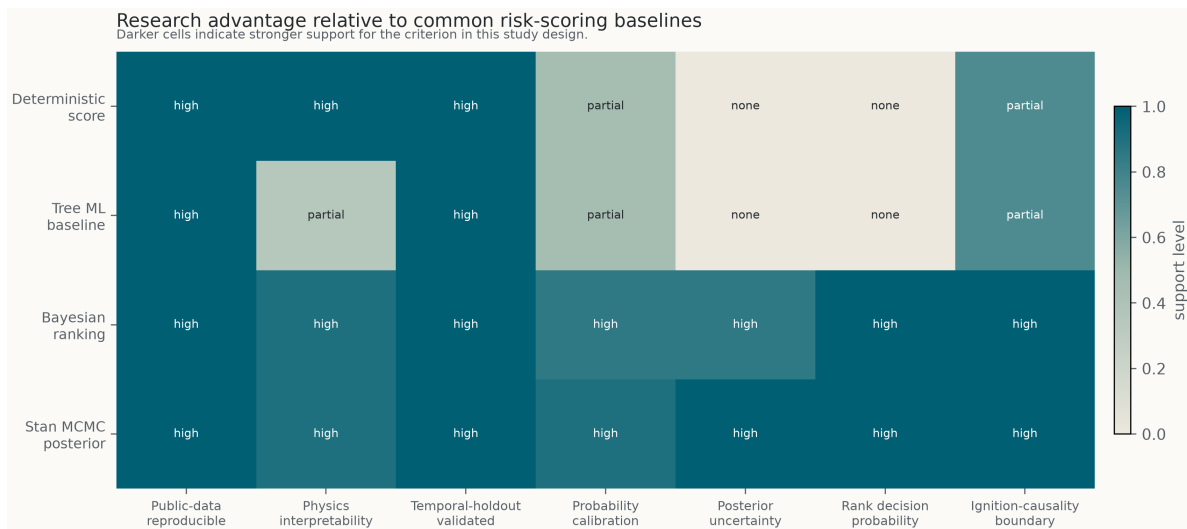


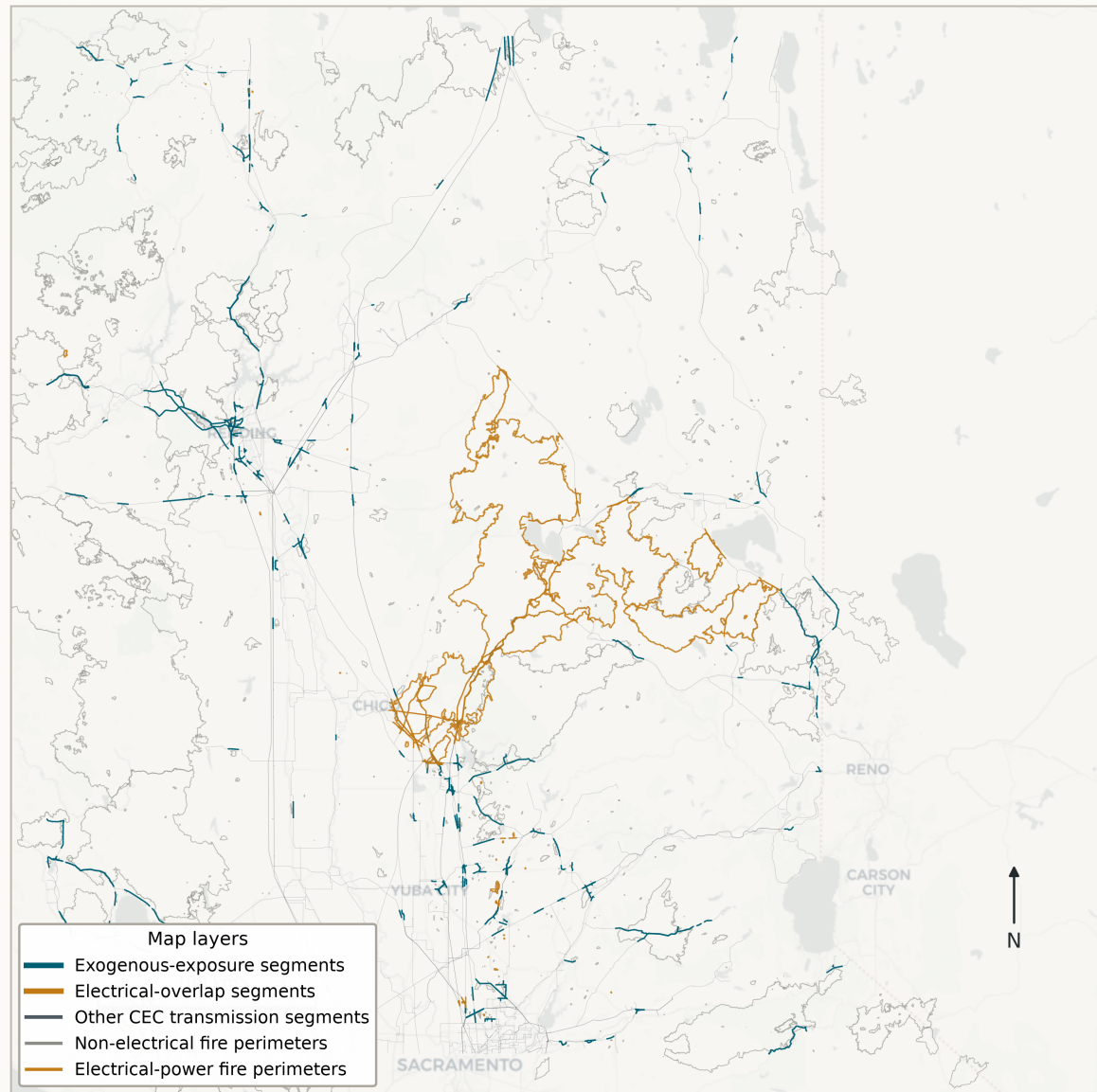
Figure 1: Positioning of the proposed public-data exposure-ranking framework.

2 Study Area and Public Data

2.1 Study Area

The case study focuses on the Northern Sierra and Southern Cascades in California. The decision units are approximately 1 km transmission-line segments. The analysis panel spans 2017–2023 and uses segment-year rows so that annual fire labels and annual fire-weather summaries are aligned. The current prototype uses a fixed 1 km segment buffer for perimeter-overlap labels; buffer sensitivity is left for follow-up validation. Figure 2 shows the study region and receptor-asset segments.

Transmission-line wildfire exposure labels in Northern California 1 km CEC line segments, 1 km buffers, CAL FIRE perimeters 2017-2023



Sources: CEC transmission lines; CAL FIRE FRAP fire perimeters; CartoDB Positron basemap.

Figure 2: Study region, transmission-line segments, and annual exposure labels.

2.2 Public Data Sources

Table 1 summarizes the public datasets and their roles. The processed panel contains 11,255 line segments and 78,785 segment-year observations. The environmental covariate layers used for quality control and feature construction are shown in Figure 3, Figure 4, and Figure 5.

Table 1: Public data sources and roles.

Dataset	Source	Spatial/temporal resolution	Years / version	Role	Notes and limitations
Transmission lines	California Energy Commission	Line geometries	Current public release	Receptor assets	External exposure receptors; TODO citation.
Fire perimeters	CAL FIRE FRAP	Perimeter polygons with cause fields	2017–2023 subset	Annual exposure labels	Exposure proxy, not damage or outage; TODO citation.
gridMET	University of Idaho gridMET	Daily, about 4 km; annual summaries	2017–2023	Weather covariates	Annual/fire-season summaries; TODO citation.
LANDFIRE	LANDFIRE LF2024	Fuel and canopy rasters	LF2024	Fuel-structure covariates	Test time-matched releases; TODO citation.
USGS 3DEP DEM	USGS 3DEP	Elevation and terrain rasters	Current public DEM	Terrain context	Not a fire-spread simulation; TODO citation.

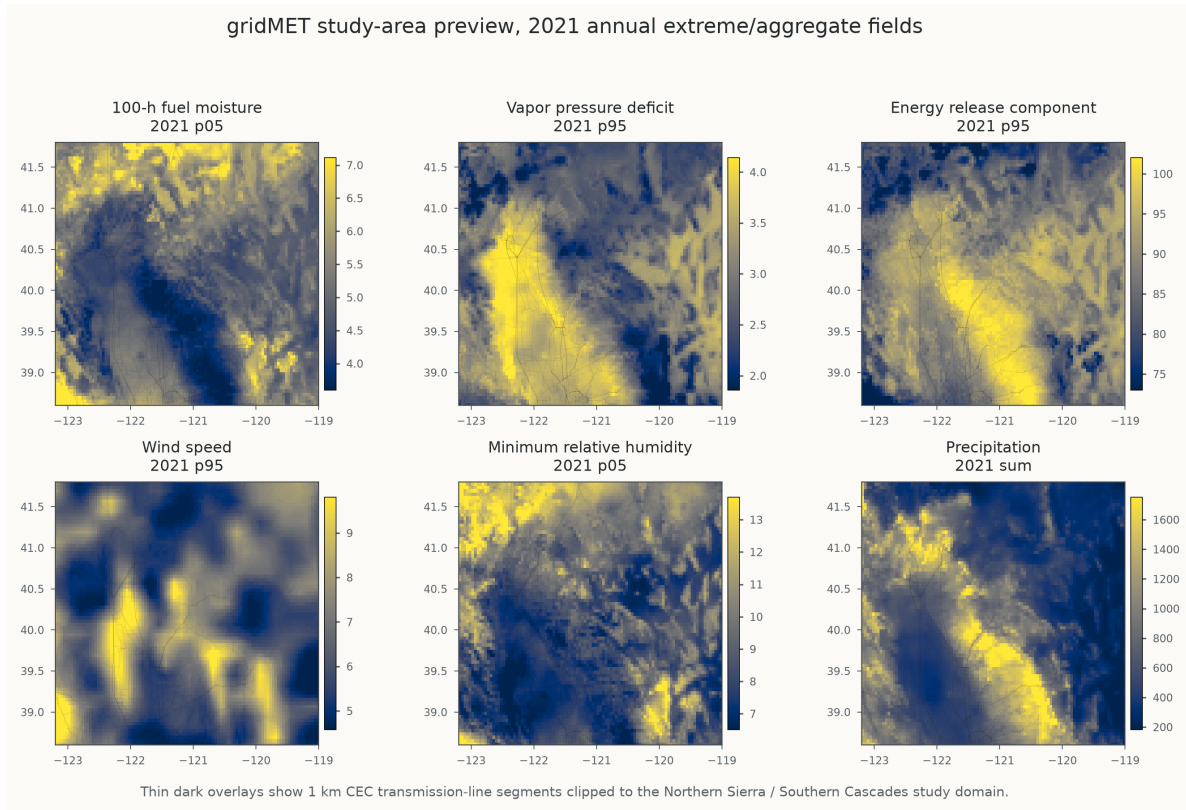


Figure 3: gridMET annual fire-weather covariate layers for 2021.

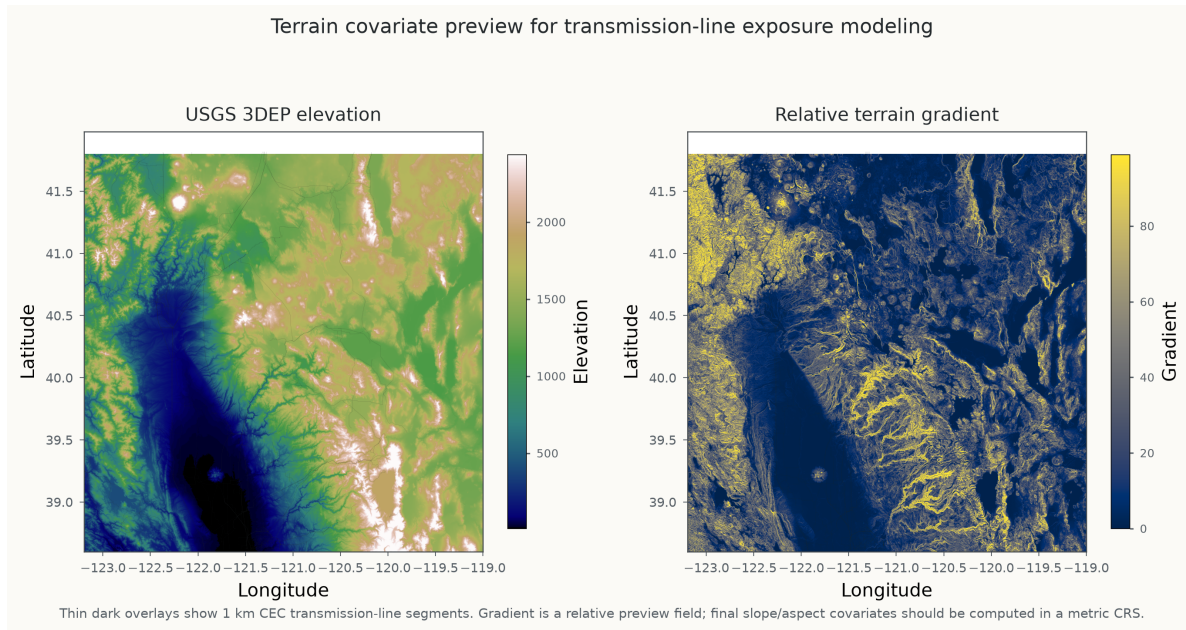


Figure 4: USGS 3DEP terrain covariate layers.

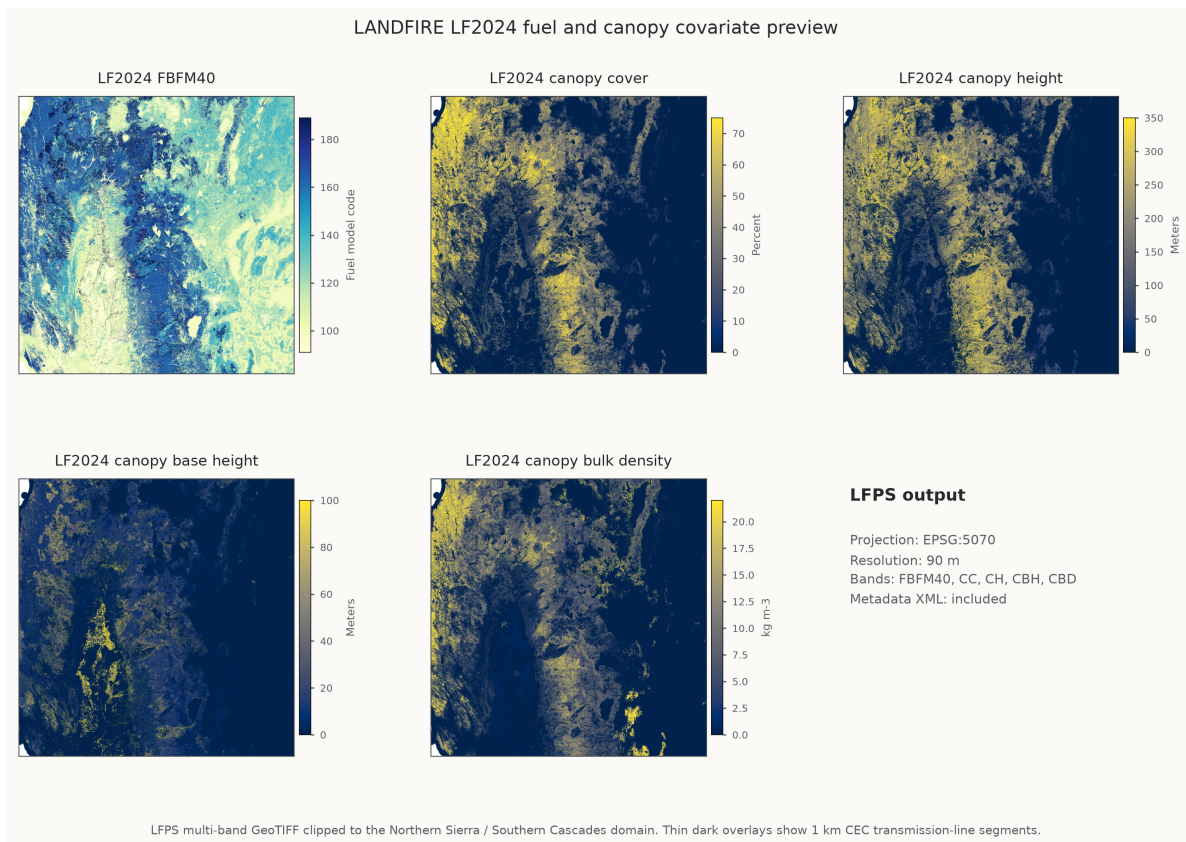


Figure 5: LANDFIRE LF2024 fuel and canopy covariate layers.

3 Label Construction

Let i index line segments and t index years. The main validation label is

$$y_{it}^{exo} = \mathbb{1}\{\text{buffer}_i \cap \text{fire perimeter}_t \neq \emptyset, \text{CAUSE} \neq 11\}, \quad (1)$$

where CAL FIRE CAUSE = 11 denotes Electrical Power. This exclusion is a label boundary choice: it is used to avoid mixing external wildfire exposure with electrical-power fire cause in the main validation target.

Four annual labels are retained. The all-fire label, y_{it}^{all} , records any buffer-perimeter overlap. The exogenous label, y_{it}^{exo} , excludes CAUSE = 11 and is the main target. The strict label, y_{it}^{strict} , excludes CAUSE in $\{11, 2\}$, where CAUSE = 2 corresponds to Equipment Use. The electrical-power overlap label, y_{it}^{elec} , records CAUSE = 11 overlap as a diagnostic only. It is not interpreted as a calibrated powerline ignition label because a perimeter-level cause code does not identify the exact ignition asset, device, or line segment.

Across 2017–2023, the panel contains 3,187 all-fire exposure positives, 2,331 exogenous exposure positives, 2,075 strict exposure positives, and 885 electrical-power overlap positives. Figure 6 maps the cause-screened labels, and Figure 7 reports annual label counts.

Cause-screened segment-year exposure labels

Labels are parallel diagnostic definitions, not mutually exclusive classes.

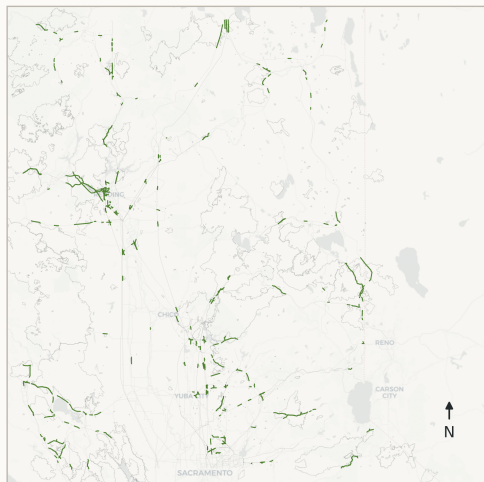
A. All fire exposure
3,187 positive segment-year rows



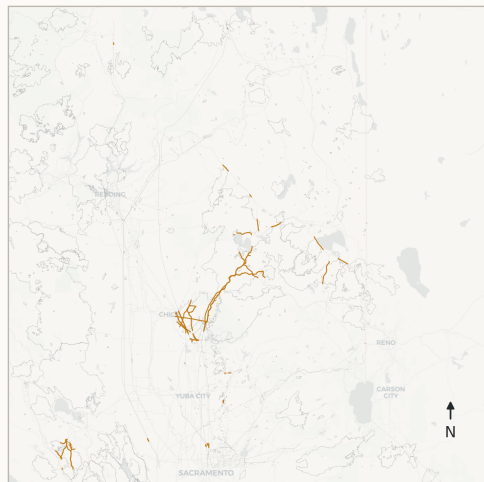
B. Exogenous exposure (CAUSE != 11)
2,331 positive segment-year rows



C. Strict exogenous (CAUSE not in {11, 2})
2,075 positive segment-year rows



D. Electrical-power overlap diagnostic
885 positive segment-year rows



— All fires — Exogenous — Strict exogenous — Electrical diagnostic

Figure 6: Cause-screened exposure labels for receptor-asset validation.

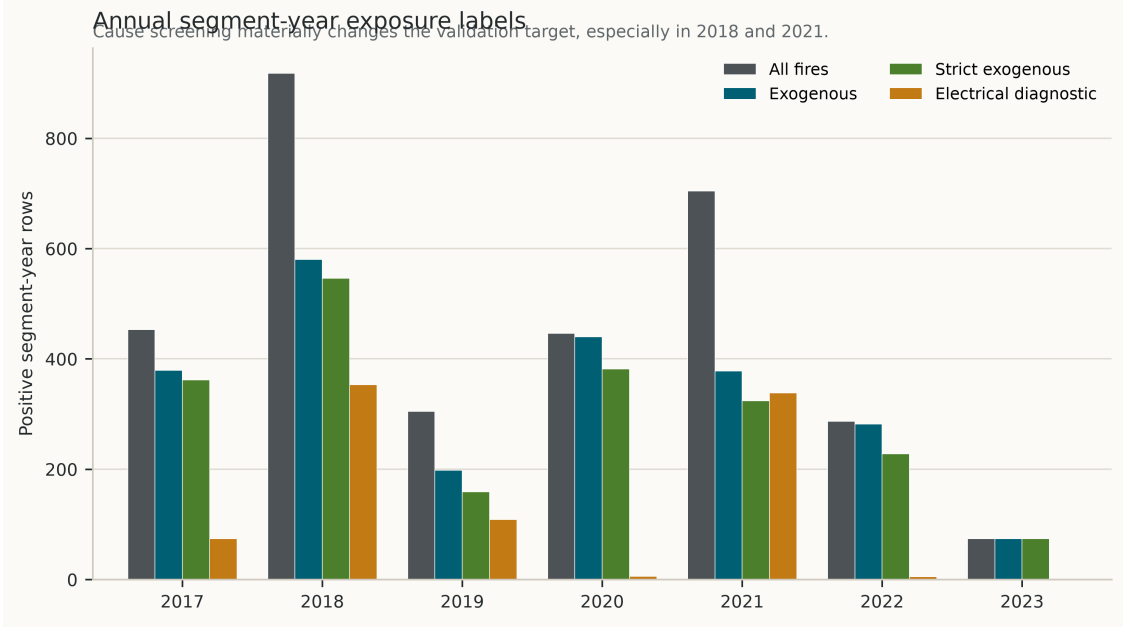


Figure 7: Annual segment-year label counts.

4 Methods

4.1 Segment Construction and Exposure Buffers

CEC transmission-line geometries are cleaned, clipped to the study region, and split into approximately 1 km decision units. Each segment is assigned an annual record for 2017–2023. The current prototype uses a fixed 1 km buffer around each segment when constructing fire-perimeter overlap labels. Future work should test 250 m, 500 m, 1000 m, and 2000 m buffers.

4.2 Environmental Covariates

The feature set combines dynamic gridMET dryness and wind summaries, static LANDFIRE fuel-structure variables, USGS terrain covariates, and simple physics-guided engineered indices. In this paper, “physics-guided” means feature engineering informed by known fire-weather, fuel, and terrain controls; it is not a physical fire-spread simulation. The selected feature core includes VPD, ERC, BI, wind speed, 100-hour and 1000-hour fuel moisture, minimum relative humidity, precipitation, elevation, canopy cover, canopy height, canopy base height, canopy bulk density, dryness index, fuel-structure index, and a dryness-by-fuel interaction. Covariate distribution diagnostics and spatial quality-control maps are shown in Figure 8 and Figure 9.

4.3 Deterministic Comparator

The deterministic comparator is a calibrated public physics-guided score based on dryness, fuel structure, and elevation. It is used as a transparent benchmark for ranking, not as the final decision model. Let $z_{k,it}$ denote standardized covariates estimated using the training period. The deterministic baseline can be written as

$$s_{it}^{det} = \text{clip}_{[0,1]} \left[a_1 z_{it}^{dry} + a_2 z_i^{fuel} + a_3 z_i^{elev} \right]. \quad (2)$$

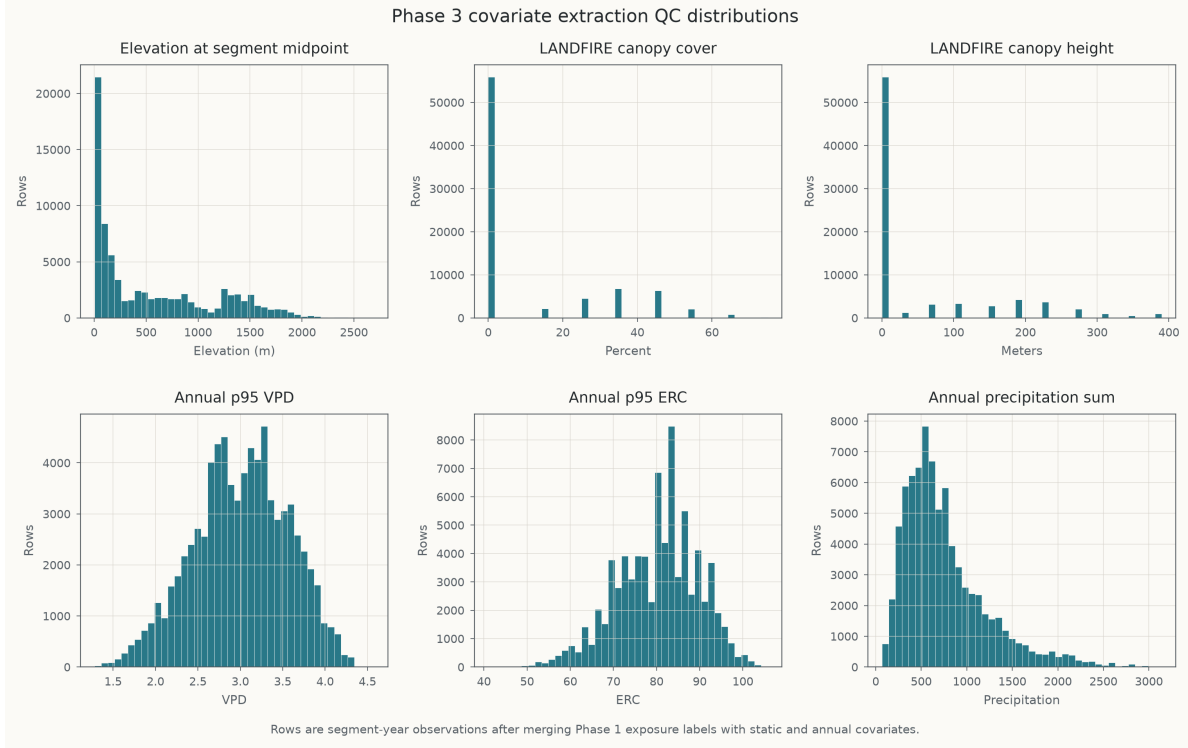


Figure 8: Covariate distribution diagnostics.

The score is subsequently calibrated to a probability scale using regularized logistic calibration. It yields a single ranking but no posterior distribution over ranks.

4.4 Bayesian Exposure Model

The main Bayesian model is a weakly regularized logistic exposure model. For segment i and year t :

$$y_{it}^{exo} \sim \text{Bernoulli}(p_{it}), \quad (3)$$

$$\eta_{it} = \alpha + \mathbf{x}_{it}^\top \boldsymbol{\beta}, \quad (4)$$

$$p_{it} = \text{logit}^{-1}(\eta_{it}), \quad (5)$$

$$\alpha \sim \mathcal{N}(0, 5^2), \quad (6)$$

$$\beta_k \sim \mathcal{N}(0, 2^2). \quad (7)$$

The posterior is

$$p(\alpha, \boldsymbol{\beta} \mid D) \propto \prod_{(i,t) \in D} p_{it}^{y_{it}} (1 - p_{it})^{1 - y_{it}} p(\alpha) \prod_k p(\beta_k). \quad (8)$$

The prototype is fit using a Laplace posterior approximation around the posterior mode and checked with an initial Stan MCMC run.

Posterior draws are indexed by $s = 1, \dots, S$:

$$(\alpha^{(s)}, \boldsymbol{\beta}^{(s)}) \sim p(\alpha, \boldsymbol{\beta} \mid D), \quad (9)$$

$$p_{it}^{(s)} = \text{logit}^{-1} \left(\alpha^{(s)} + \mathbf{x}_{it}^\top \boldsymbol{\beta}^{(s)} \right). \quad (10)$$

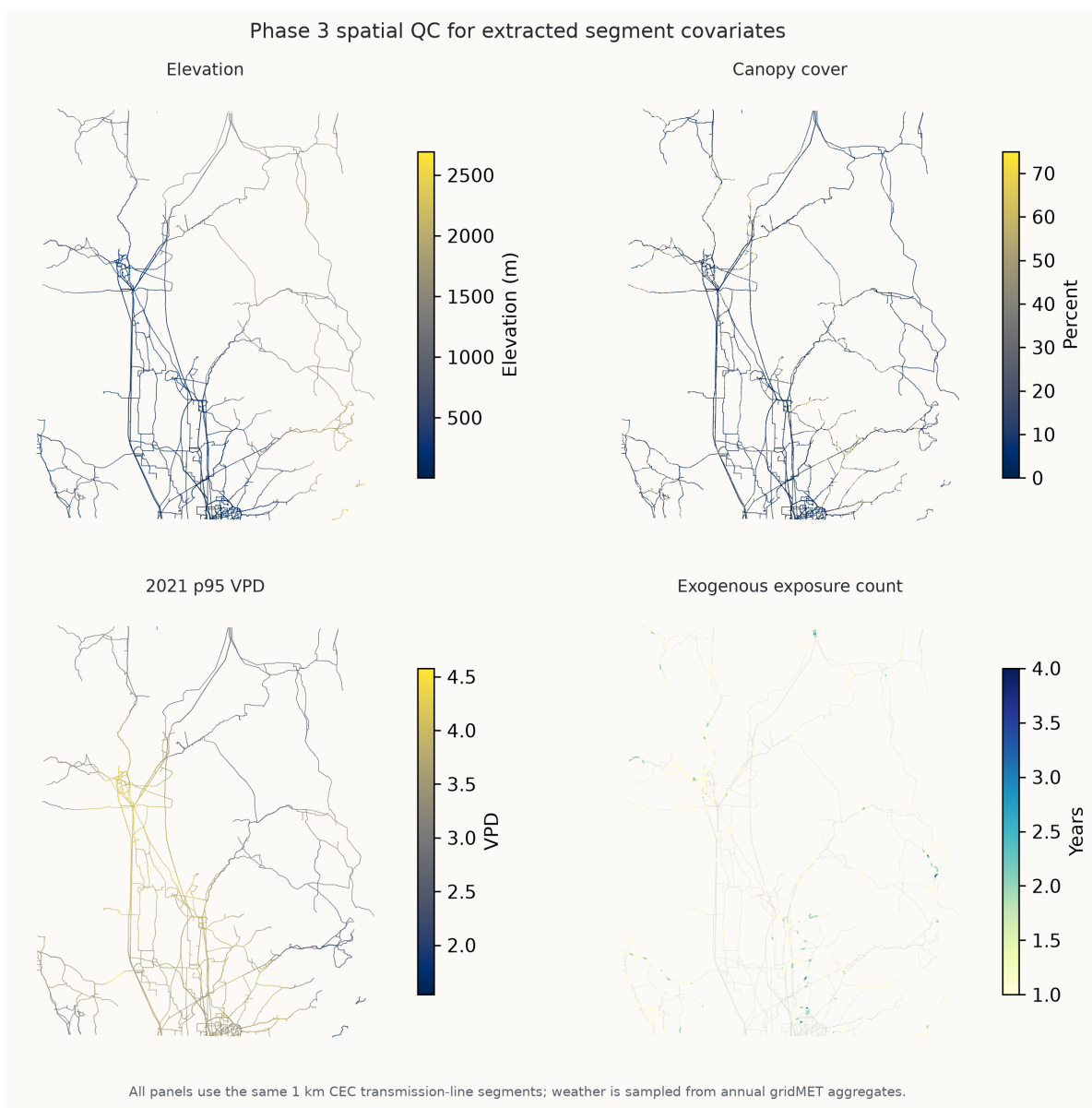


Figure 9: Spatial covariate quality-control maps.

For segment-level decisions over the holdout period $T^\star$, posterior segment exposure is aggregated as

$$\bar{p}_i^{(s)} = \frac{1}{|T^\star|} \sum_{t \in T^\star} p_{it}^{(s)}. \quad (11)$$

The decision quantities are:

$$q_i = P(\bar{p}_i > \tau \mid D) \approx \frac{1}{S} \sum_{s=1}^S \mathbb{1}\{\bar{p}_i^{(s)} > \tau\}, \quad (12)$$

$$\rho_i = P(\text{rank}(\bar{p}_i) \leq K \mid D) \approx \frac{1}{S} \sum_{s=1}^S \mathbb{1}\{\text{rank}(\bar{p}_i^{(s)}) \leq K\}, \quad (13)$$

where τ is the training prevalence and K is the top-decile segment count. Figure 10 summarizes this workflow.

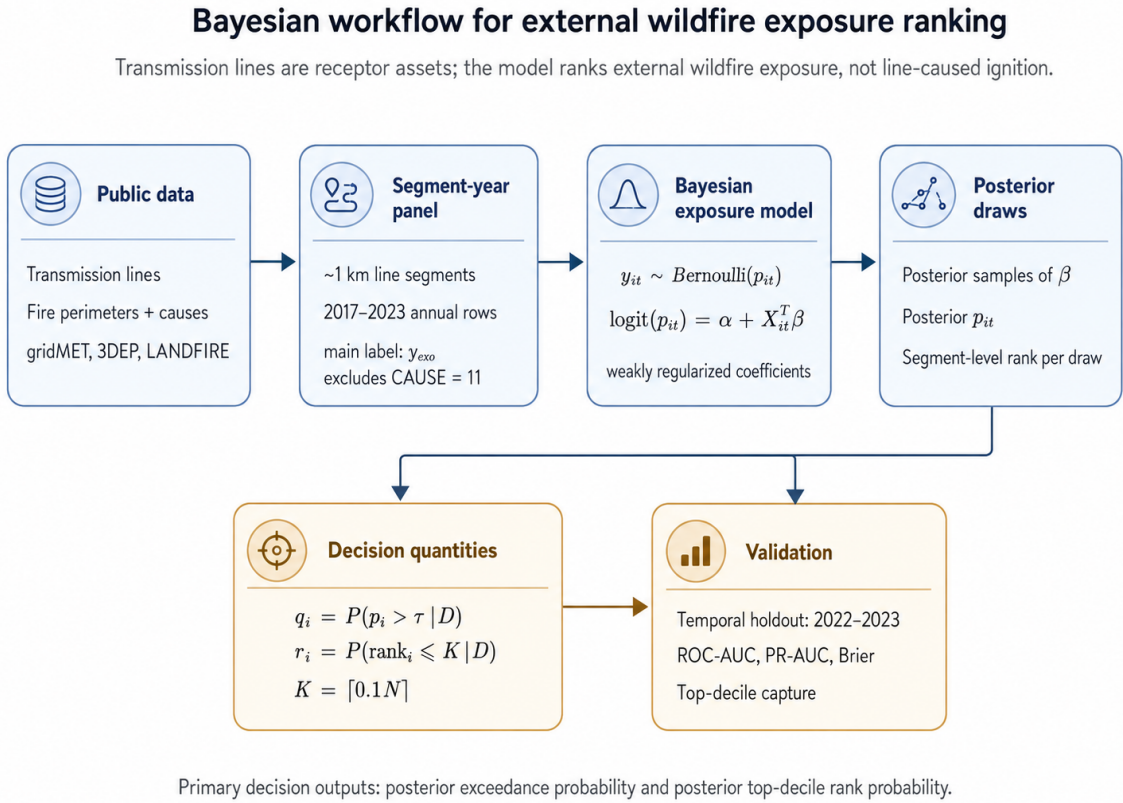


Figure 10: Bayesian workflow for posterior exposure and rank probabilities.

4.5 Validation Design

Models are trained on 2017–2021 and evaluated on 2022–2023. The holdout set contains 22,510 segment-year rows and 356 exogenous exposure positives, giving a holdout prevalence of 1.58%. This rare-event baseline is important for interpreting PR-AUC. Metrics include ROC-AUC, PR-AUC, Brier score, row-level top-decile capture, and segment-level top-decile capture.

5 Results

5.1 Temporal Holdout Performance

Table 2 compares the deterministic score, Bayesian Laplace model, Stan MCMC pilot, and machine-learning baselines. The Bayesian and regularized logistic L2 models have nearly identical ROC-AUC and PR-AUC. The Bayesian contribution is therefore not a claim of broad classification superiority; it is the posterior decision layer shown in Figure 12, Figure 13, and Figure 17. Relative to the deterministic calibrated score, the Bayesian ranking improves rare-event decision concentration in this temporal holdout, summarized in Figure 11.

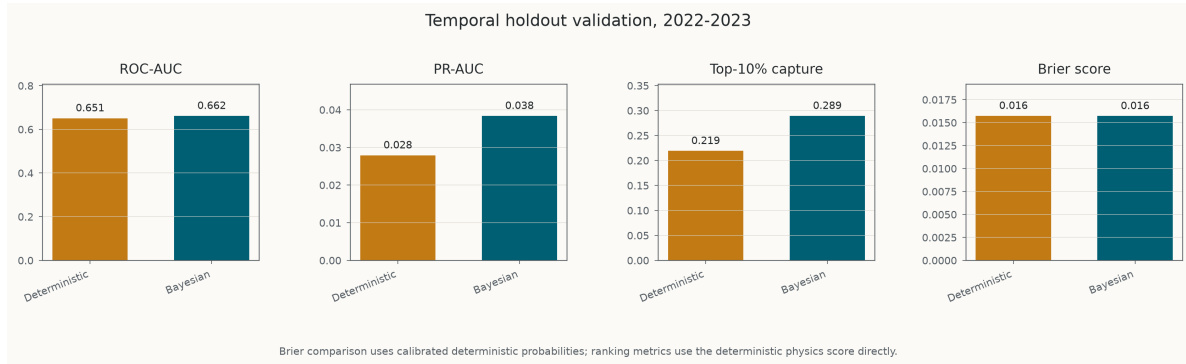


Figure 11: Temporal-holdout validation for deterministic and Bayesian ranking.

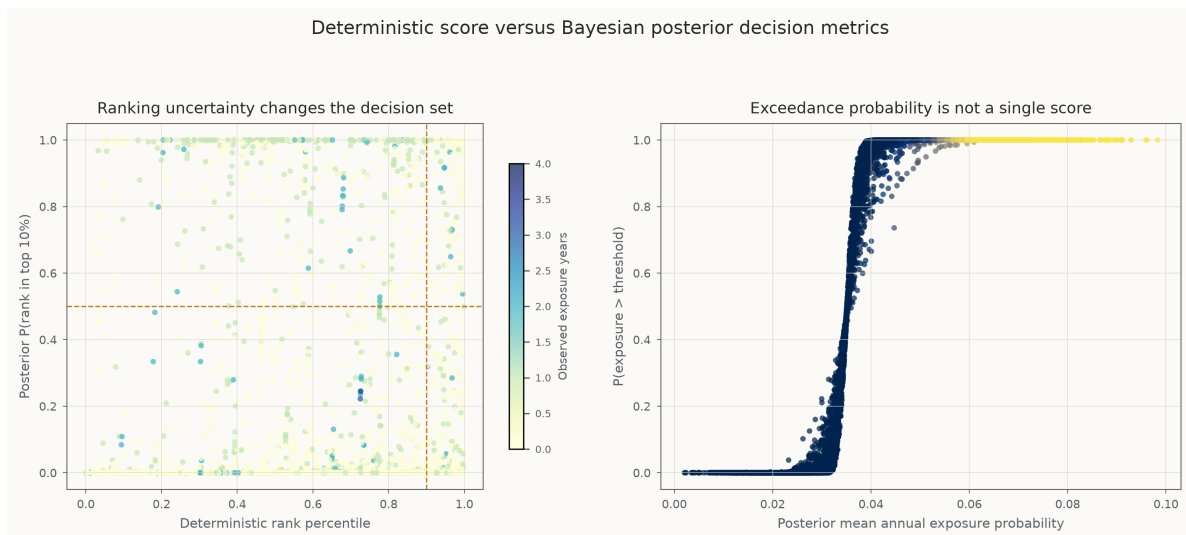


Figure 12: Deterministic score and Bayesian posterior decision metrics.

5.2 Feature Ablation and Baseline Checks

Table 3 and Figure 14 summarize feature-set ablations. Dynamic gridMET weather summaries carry much of the temporal-holdout discrimination. LANDFIRE fuel variables are useful context variables but do not by themselves support strong holdout discrimination in this prototype. Figure 15 compares deterministic, Bayesian, and ML baselines, and Figure 16 reports the calibration

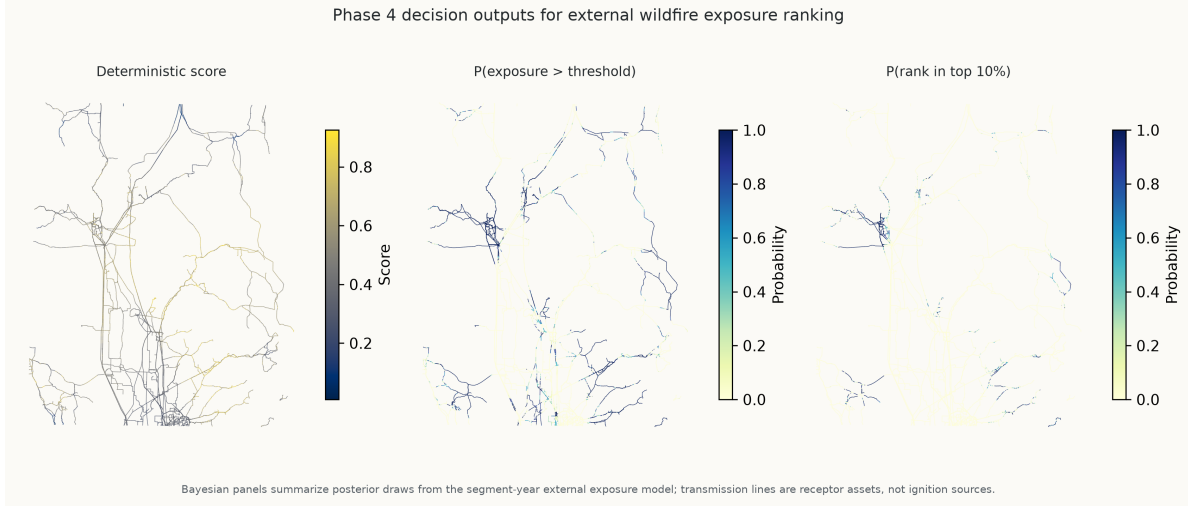


Figure 13: Posterior decision metrics mapped to transmission-line segments.

Table 2: Temporal-holdout comparison for 2022–2023 exogenous exposure.

Model	ROC-AUC	PR-AUC	Brier	Row top 10%	Segment top 10%
ML logistic L2	0.663	0.039	0.015713	0.289	0.216
Bayesian Laplace	0.662	0.038	0.015718	0.289	0.216
Stan MCMC pilot	0.662	0.038	0.015710	0.289	0.216
Deterministic calibrated	0.651	0.028	0.015734	0.219	0.118
Balanced extra trees	0.617	0.022	0.056091	0.180	0.104
Histogram gradient boosting	0.553	0.019	0.016647	0.132	0.149
Balanced random forest	0.568	0.018	0.034539	0.107	0.152

diagnostic.

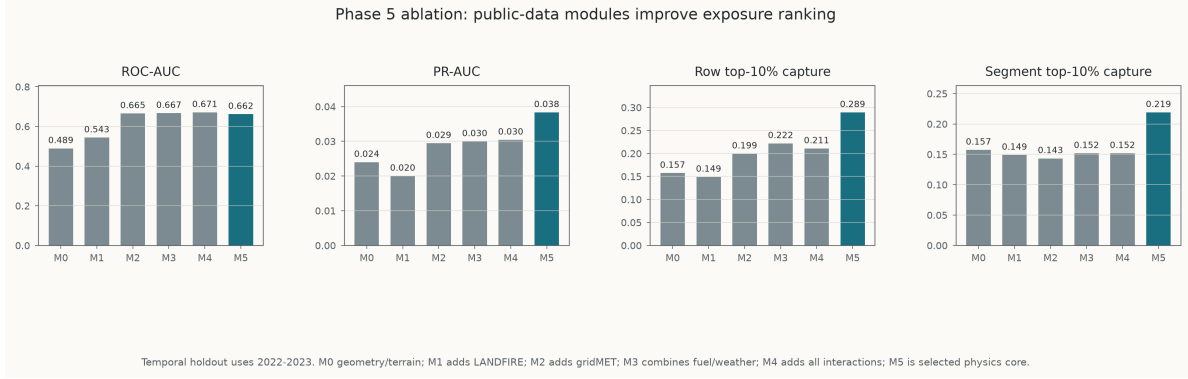


Figure 14: Feature-set ablation under temporal holdout.

Table 3: Feature-set ablation under 2022–2023 temporal holdout.

Model	ROC-AUC	PR-AUC	Brier	Row top 10%	Segment top 10%
M0 geometry + terrain	0.489	0.024	0.016052	0.157	0.157
M1 + LANDFIRE fuel	0.543	0.020	0.016076	0.149	0.149
M2 + gridMET weather	0.665	0.029	0.015822	0.199	0.143
M3 fuel + weather	0.667	0.030	0.015821	0.222	0.152
M4 physics interaction	0.671	0.030	0.015820	0.211	0.152
M5 selected physics core	0.662	0.038	0.015723	0.289	0.219

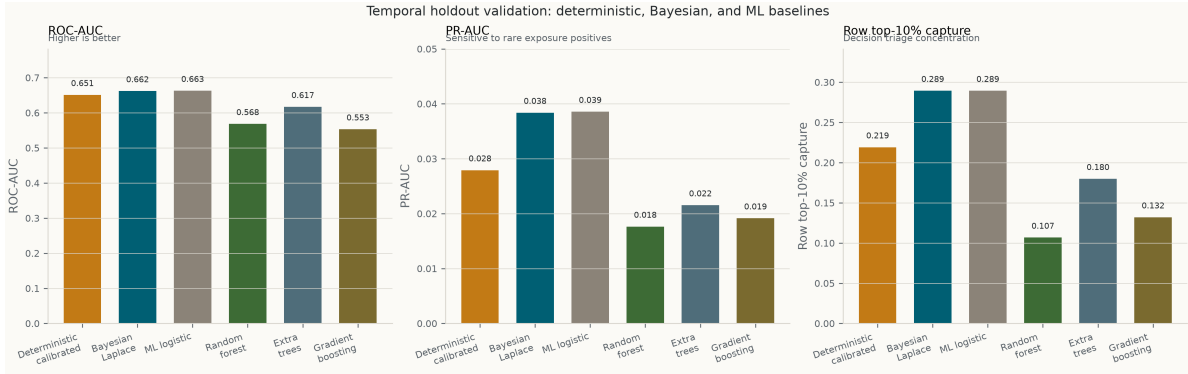


Figure 15: Deterministic, Bayesian, and machine-learning baseline comparison.

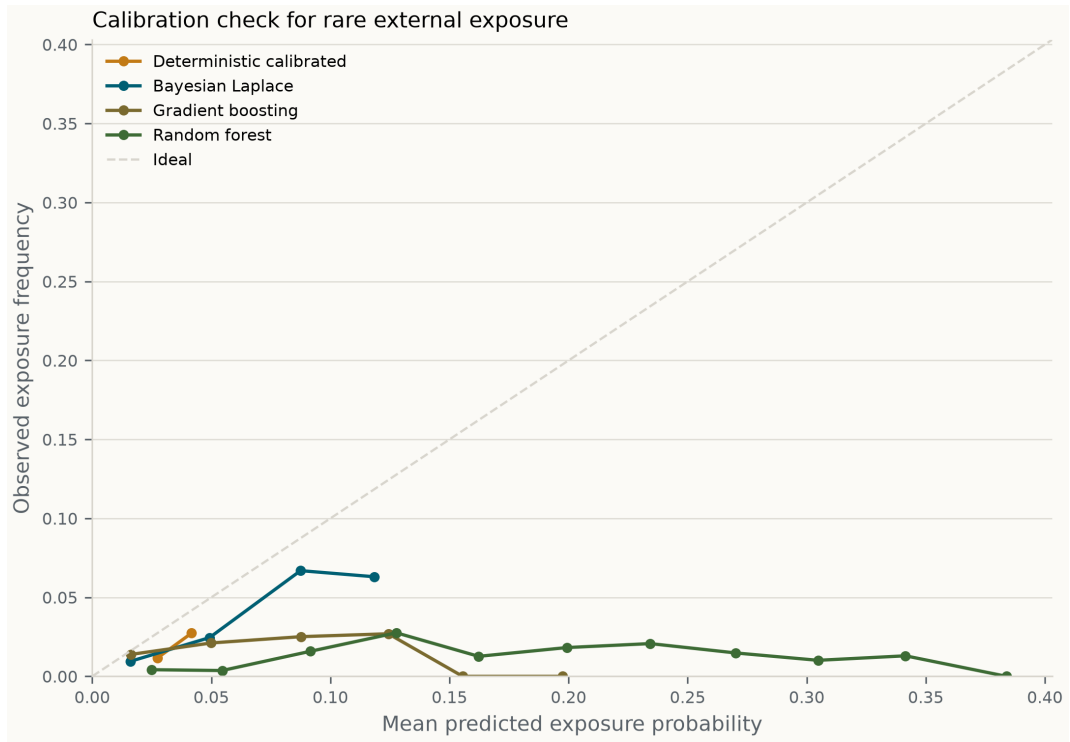


Figure 16: Calibration diagnostic for rare external exposure probabilities.



Figure 17: Full-region and zoomed maps of posterior decision metrics.

6 Discussion

The main result is a bounded one: public data can support a reproducible, screening-level framework for external wildfire exposure ranking of transmission-line receptor assets. The output should be read as a ranked exposure screen, not as ignition causality, equipment condition, outage likelihood, damage, or responsibility.

The temporal-holdout results do not show that Bayesian modeling is universally more accurate than all machine-learning alternatives. The regularized logistic L2 baseline and the Bayesian Laplace model have nearly identical discrimination in Table 2. The Bayesian advantage in this paper is decision-facing: it converts the fitted exposure model into posterior exceedance probability and posterior top-decile rank probability. These quantities are more aligned with rare-event triage than a single deterministic score because they represent uncertainty in threshold and ranking decisions.

The deterministic comparator remains useful because it is transparent and easy to audit. Its limitation is that it produces one calibrated score and therefore cannot express posterior rank uncertainty. Tree-based ML baselines are more flexible, but in this temporal split they do not improve rare-event concentration relative to the selected physics-guided linear core. This result should be interpreted as evidence for cautious feature design and temporal validation, not as a general rejection of ML models.

Existing wildfire risk tools often target different questions. Powerline ignition and equipment-failure models generally require private utility data and are aimed at grid-to-fire causality. Fire-spread simulators can represent richer physical dynamics but require scenario assumptions and are not automatically calibrated for public-data segment ranking. The present framework occupies a narrower space: fire-to-grid external exposure ranking using public data and a posterior decision layer.

7 Limitations

Several limitations are important for an EarthArXiv preprint. First, the pipeline estimates external exposure only. It does not estimate ignition causality, equipment failure, outage probability, damage, or legal responsibility. Second, the CAL FIRE cause field is a perimeter-level attribute; electrical-power overlap is therefore retained only as a diagnostic label. Third, the current prototype uses LANDFIRE LF2024 static fuel layers for a 2017–2023 panel. This may create temporal mismatch with earlier years and should be tested against time-matched LANDFIRE releases in a journal version. Fourth, the label construction uses a fixed 1 km buffer. Buffer sensitivity should compare 250 m, 500 m, 1000 m, and 2000 m definitions. Fifth, temporal holdout is not a substitute for spatial block validation. Finally, the Stan run is an initial MCMC check; full Stan diagnostics, longer chains, convergence summaries, and posterior predictive checks are still needed.

8 Conclusion

This paper presents a public-data framework for external wildfire exposure ranking of transmission-line segments treated as receptor assets. The workflow builds a 2017–2023 segment-year panel, uses cause-screened exposure labels, and compares deterministic, Bayesian, and machine-learning baselines under a 2022–2023 temporal holdout. The Bayesian and regularized logistic models show similar discrimination, while the Bayesian formulation adds posterior exceedance and posterior top-decile rank probabilities. These posterior decision quantities are the main methodological

contribution for rare-event exposure triage. The framework should not be interpreted as a powerline ignition, equipment-failure, outage, damage, or responsibility model.

A Reproducibility Notes

All data processing, feature construction, model fitting, validation, and figure generation are scripted in the project repository. The public-data workflow is organized by phases: asset and label construction, environmental covariate preparation, model-table construction, Bayesian ranking, ablation, ML/Stan validation, and manuscript result generation. Manuscript tables and figures are generated from scripted outputs rather than manually transcribed. Public-source citations remain marked as TODO placeholders until a bibliography is added.